Aspect-Based Sentiment Analysis for Automated Product Review Analysis Using Natural Language Processing

Project presentation

The problem

The screen is gorgeous and the keyboard feels great, but the battery life is terrible.

- Traditional sentiment analysis: one label for the whole review
- What a seller needs: screen positive, keyboard positive, battery life negative
- Reading every review by hand does not scale

Objectives

1. Extract product aspects with a learnt model, with no hard-coded aspect list
2. Classify the sentiment towards each aspect
3. Measure both on a standard benchmark and against a non-neural baseline
4. Deliver a working web application

Dataset: SemEval-2014 Task 4

- Laptop and restaurant review sentences, annotated by people
- Aspect terms with character offsets and polarity
- Official gold test set used unchanged; duplicates and train/test overlaps removed

Domain	Training	Validation	Test
laptops	2723	303	800
restaurants	2726	303	800

Method: two stages

```
review -> clean, split, tokenise
       -> BERT token classifier     (BIO tags: which words are aspects?)
       -> BERT sentence-pair model  ([CLS] sentence [SEP] aspect [SEP] -> sentiment)
       -> aspect-sentiment map + overall sentiment
```

- Both start from bert-base-uncased, fine-tuned end to end
- A third model gives the traditional single label, for comparison

Why BIO tags

```
The   battery    life       is  great  but  the  screen     is  dim
O     B-ASPECT   I-ASPECT   O   O      O    O    B-ASPECT   O   O
```

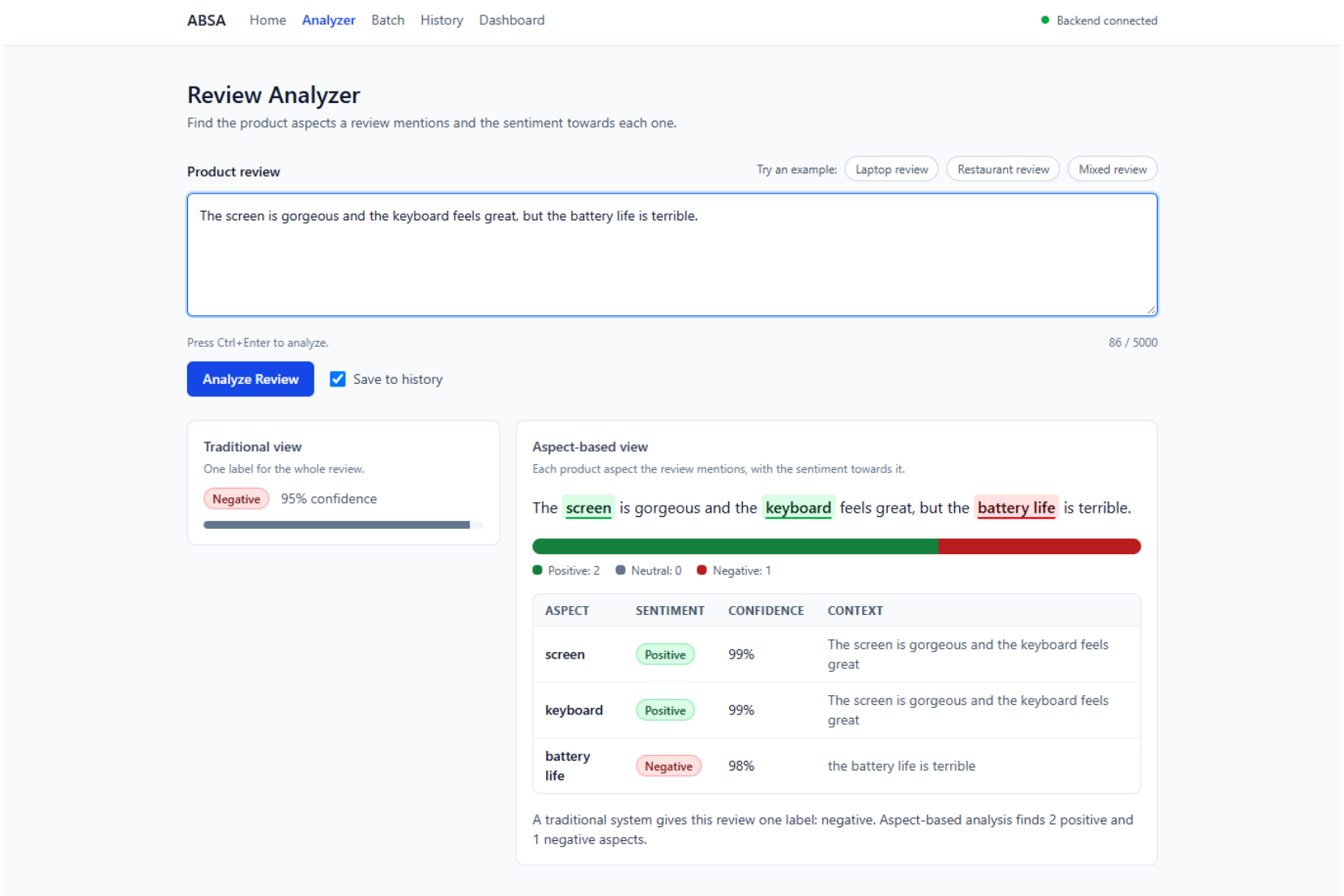
- Multi-word aspects are one unit
- Adjacent aspects stay separate
- No stop-word removal: BERT needs "not" and "but"

System

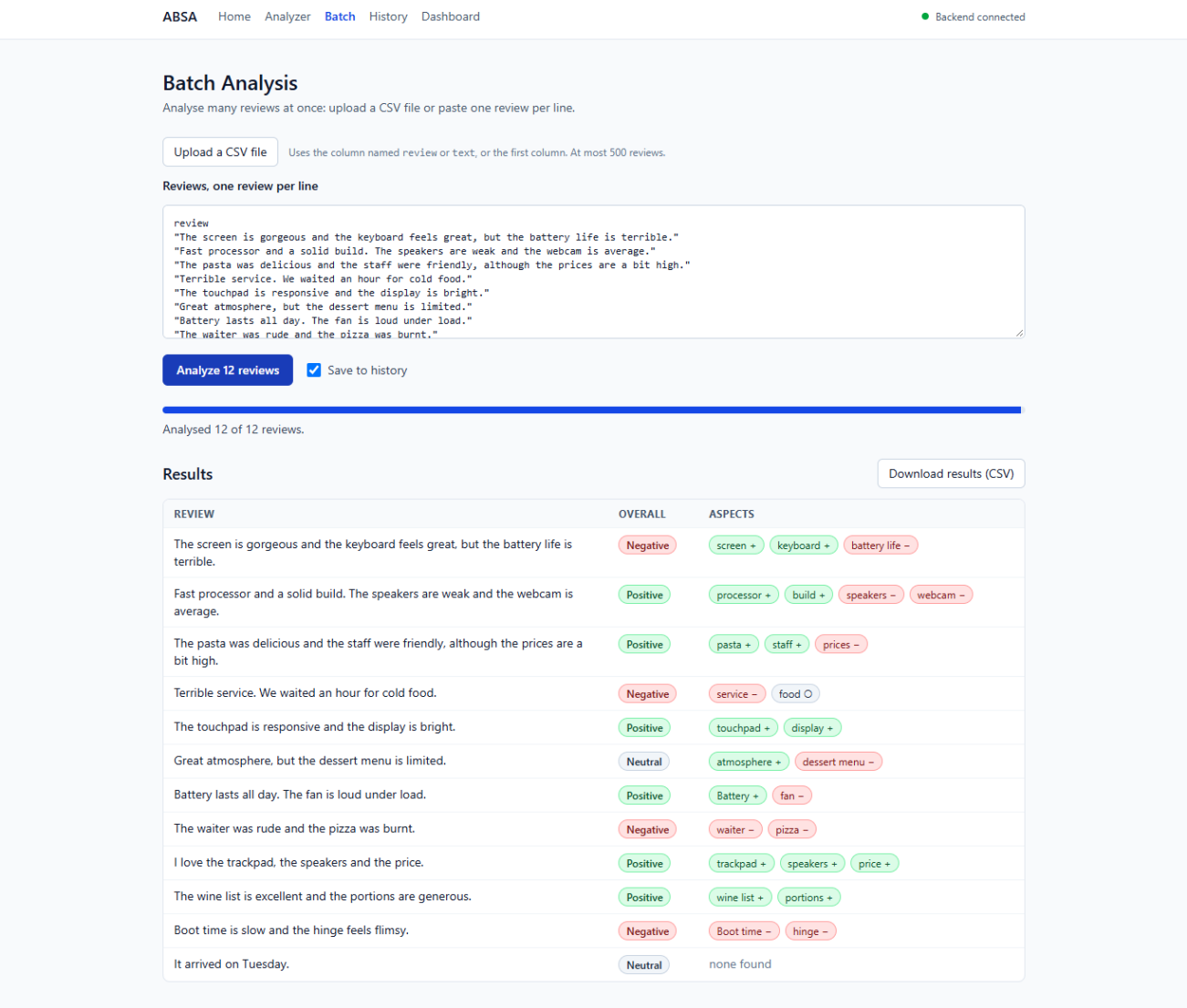
```
React + TypeScript  ->  FastAPI  ->  ABSA pipeline (3 BERT models)
                                 ->  SQLite (review history)
```

- Analyzer, batch CSV analysis, history, dashboard
- Every metric and document is regenerated by script
- Automated tests for backend and frontend

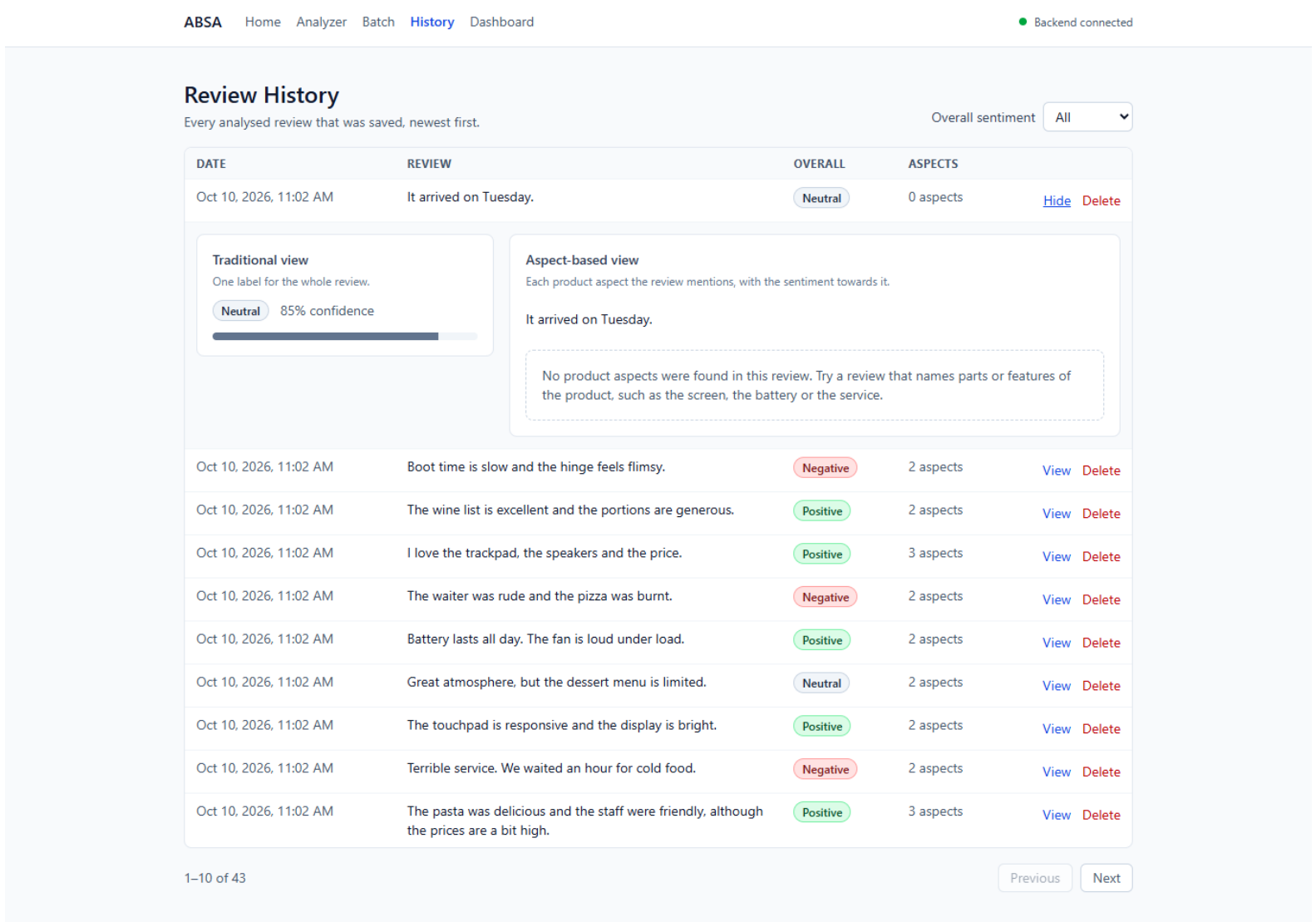
Application: analyzing one review



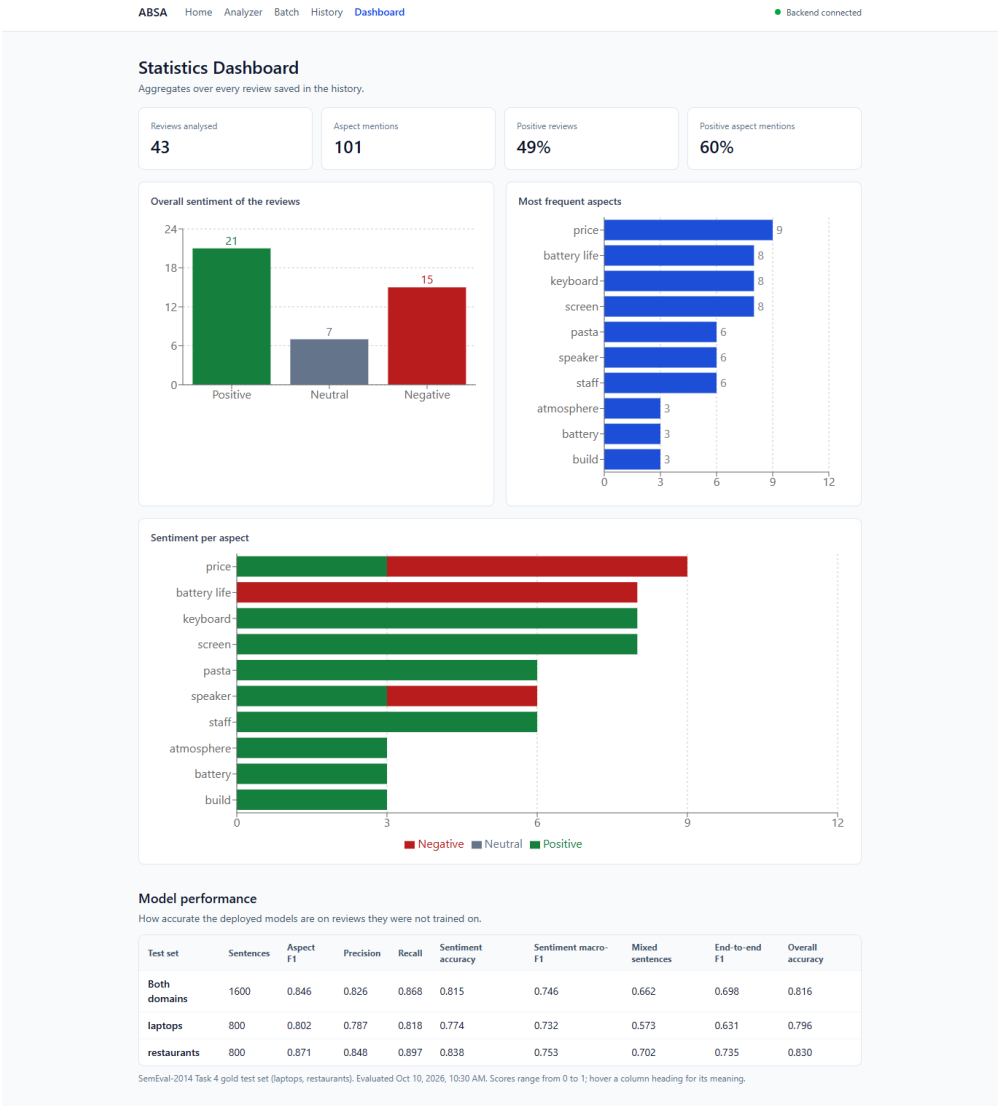
Application: batch analysis of a CSV file



Application: review history



Application: dashboard and model performance



Results on the test sets

Metric	Both domains	Laptops	Restaurants
Aspect extraction F1	0.846	0.802	0.871
Aspect sentiment accuracy	0.815	0.774	0.838
Aspect sentiment macro-F1	0.746	0.732	0.753
End-to-end F1	0.698	0.631	0.735

Exact span match; end-to-end requires aspect and sentiment both right.

What BERT adds

Baseline: noun phrases as aspects, opinion-word lexicon for sentiment. Same data, same metrics.

Metric	Baseline	BERT
Aspect extraction F1	0.376	0.846
Aspect sentiment accuracy	0.663	0.815
Aspect sentiment macro-F1	0.616	0.746
End-to-end F1	0.253	0.698

Transfer to an unseen domain

Restaurant test set: an aspect extractor trained on laptops only against the one trained on both domains.

Metric	Trained on laptops only	Trained on both
Aspect extraction precision	0.892	0.848
Aspect extraction recall	0.234	0.897
Aspect extraction F1	0.370	0.871

Where it still fails

- Aspects in sentences with one sentiment: accuracy 0.851
- Aspects in sentences that mix sentiments: accuracy 0.662
- Negative class F1: 0.749
- Neutral class F1: 0.581
- Positive class F1: 0.906
- Only explicit aspects are found

Limitations

- Training data: two domains, from 2014
- One training run, one seed, no confidence intervals
- Overall-sentiment labels are derived, not annotated
- Confidence values are not calibrated

Demo

1. Analyze a laptop review and a restaurant review
2. A review with no aspect
3. Batch: upload a CSV, export the results
4. History and dashboard

Script: docs/demo.md

Conclusion and future work

- A complete, tested ABSA system with reproducible evaluation
- Next: more data with contrasting aspects, aspect position in the input, several seeds, implicit aspects, more product categories

Thank you. Questions?